The University of Texas at Austin
In-Class Work 17

M339D Introduction to Financial Mathematics

November 03, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Log-Normal Stock Prices.

Problem 17.1: *The current price of a non-dividend-paying stock is \$80 per share. Under the risk-neutral probability measure, its mean rate of return is 12% and its volatility is 30%.*

Let $R(0,t)$ *denote the realized return of this stock over the time period* $[0,t]$ *for any* $t>0$. *Calculate* $\mathbb{E}^*[R(0,2)]$.

Problem 17.2: *A stock is valued at* $\$75.00$. *The continuously compounded, risk-free interest rate is* 10.0% *and the standard deviation of annualized returns is* 25.0%. *If the stock is lognormally distributed, what is the expected stock price after* 2 *years under the risk-neutral probability measure?*

Problem 17.3: *A non-dividend-paying stock is valued at \$55.00 per share. Its standard deviation of annualized returns is given to be 22.0%. The continuously compounded, risk-free interest rate is 12%. If the stock price is modeled using the lognormal distribution (as discussed in class), what is the median of the stock price in 3 years under the risk-neutral probability measure?*

Problem 17.4: *Assume that the stock price is modeled using the lognormal distribution. The stock pays no dividends. Under $\mathbb{P}^*$, the annual mean rate of return on the stock is given to be 12%. Also under $\mathbb{P}^*$, the median time$-t$ stock price is evaluated to be $S(0)e^{0.1t}$. What is the volatility parameter of this stock price?*

Problem 17.5: *The current stock price is $\$100$ per share. The stock price at any time $t > 0$ is modeled using the lognormal distribution. Assume that the continuously compounded risk-free interest rate equals 8%. Let its volatility equal 20%.*

Find the value t^* at which the median stock price equals $\$120$, under the risk-neutral probability measure.

Problem 17.6: *The volatility of the price of a non-dividend-paying stock is* 0.2*. The stock price is modeled using a log-normal distribution. Under* $\mathbb{P}^*$*, the expected time—*2 *stock price is \$120. What is the median of the time—*2 *stock price under* $\mathbb{P}^*$*?*